Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x,\mu,t;c) = ce^{-(1-c)v\Sigma_t t}\psi(cx,\mu,ct;1) \tag{1}$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx,\mu,ct;1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t}\tilde{\psi}$ satisfies the same equation as $\psi(x,\mu,t;c)$.

1.1 The equation for $\tilde{\psi}$

Notation: We will denote by $\Psi(X,\mu,T) \equiv \psi(X,\mu,T;1)$ the $c=1$ solution written in its own variables, and by Ψ_T and Ψ_X its partial derivatives with respect to them, so that $\tilde{\psi}(x,\mu,t) = \Psi(cx,\mu,ct)$ with $X = cx$ and $T = ct$. **This notation is used to avoid confusion when using the chain rule, as will be done immediately.** First, Ψ satisfies the $c=1$ equation in the variables X and T:

$$\frac{1}{v}\Psi_T + \mu\Psi_X + \Sigma_t\Psi = \frac{\Sigma_t}{2}\int_{-1}^{1}\Psi d\mu' \tag{2}$$

which after applying the chain rule, $\Psi_T = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial t}$ and $\Psi_X = \frac{1}{c}\frac{\partial\tilde{\psi}}{\partial x}$, and multiplying by c, becomes:

$$\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi} = \frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' \tag{3}$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x,\mu,t;c)$ except that it is removed at the rate $c\Sigma_t$ instead of Σ_t.

1.2 Proving the identity

In this part we are going to simply subtitute the RHS of **??** into the LHS of the equation for $\psi(x,\mu,t;c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we get:

$$\text{LHS} = \frac{1}{v}\left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t}\tilde{\psi} + c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial t}\right] + \mu c\,e^{-(1-c)v\Sigma_t t}\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t c\,e^{-(1-c)v\Sigma_t t}\tilde{\psi} \tag{4}$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + \Sigma_t\tilde{\psi} + (c-1)\Sigma_t\tilde{\psi}\right] \tag{5}$$

$$= ce^{-(1-c)v\Sigma_t t}\left[\frac{1}{v}\frac{\partial\tilde{\psi}}{\partial t} + \mu\frac{\partial\tilde{\psi}}{\partial x} + c\Sigma_t\tilde{\psi}\right] \tag{6}$$

$$= ce^{-(1-c)v\Sigma_t t}\frac{c\Sigma_t}{2}\int_{-1}^{1}\tilde{\psi}d\mu' = \frac{c\Sigma_t}{2}\int_{-1}^{1}ce^{-(1-c)v\Sigma_t t}\tilde{\psi}d\mu' \tag{7}$$

where **??** was used in the last line, and this is exactly the RHS of the equation for $\psi(x,\mu,t;c)$.

2 Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi\int_{|x|}^{\infty}\phi_{\text{pt}}(r)rdr \tag{8}$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r,t)$ for $c=1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pl}}(x,t) = 2\pi\int_{|x|}^{\infty}\frac{e^{-v\Sigma_t t}}{4\pi r^2}\delta(r-vt)rdr + 2\pi\int_{|x|}^{\infty}\frac{\left(1-r^2/v^2t^2\right)^{1/8}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}e^{-v\Sigma_t t}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)\Theta(vt-r)rdr \tag{9}$$

$$= \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}}{\left(4\pi vt/(3\Sigma_t)\right)^{3/2}}\int_{|x|}^{vt}\left(1-\frac{r^2}{v^2t^2}\right)^{1/8}G\left(v\Sigma_t t\left[1-\frac{r^2}{v^2t^2}\right]^{3/4}\right)rdr \tag{10}$$

$$\overset{\xi=r/vt}{=} \frac{e^{-v\Sigma_t t}}{2vt}\Theta(vt-|x|) + \Theta(vt-|x|)\frac{2\pi e^{-v\Sigma_t t}(vt)^{1/2}}{\left(4\pi/(3\Sigma_t)\right)^{3/2}}\int_{\frac{|x|}{vt}}^{1}\left(1-\xi^2\right)^{1/8}G\left(a\left[1-\xi^2\right]^{3/4}\right)\xi d\xi \tag{11}$$

$$\overset{\substack{u=1-\xi^2\\ w=au^{3/4}}}{=} \frac{e^{-v\Sigma_t t}}{2vt}\left[1+\sqrt{\frac{3}{\pi}}\int_0^{w_0}\sqrt{w}G(w)dw\right]\Theta(vt-|x|) \tag{12}$$

with $w_0 = a\left(1-\frac{x^2}{(vt)^2}\right)^{3/4}$, the powers combining as $\frac{1}{6}+\frac{1}{3}=\frac{1}{2}$ and the Σ_t dependence cancelling in the last step (I used the assistance of AI in this algebric derivation). Finally, with the interpolation formula of Paasschens, $G(w) = e^w\sqrt{1+b/w}$ with $b = 2.026$, the integral has a closed form:

$$\int_0^{w_0}e^w\sqrt{w+b}\,dw = e^{w_0}\sqrt{w_0+b} - \sqrt{b} - \frac{\sqrt{\pi}}{2}e^{-b}\left[\text{erfi}\left(\sqrt{w_0+b}\right) - \text{erfi}\left(\sqrt{b}\right)\right] \tag{13}$$

which **closes the derivation of the planar-source scalar flux for** $c=1$**.** Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c=1$.

3 Question 3 - Classical and asymptotic diffusion against the exact solution

Throughout $\Sigma_t = v = 1$ and the source is the plane pulse $Q(x,t) = \delta(x)\delta(t)$ — a pulse in time as well as in space, since Paasschens takes $S = \delta(\mathbf{r})\delta(t)$ (his Eq. 4) and the uncollided term $\propto \delta(r - vt)$ is its signature. Figures are generated by `src/homework2`; the derivations behind the code are in `src/homework2/explanations/`.

3.1 The three solutions compared

The **exact** solution is the planar flux of Question 2, carried to general c by the identity of Question 1:

$$\phi(x,t;1) = \frac{e^{-t}}{2t}\left[1 + \sqrt{\frac{3}{\pi}}\int_0^{w_0} \sqrt{w}\, G(w)\, dw\right]\Theta(t - |x|), \qquad w_0 = t\left(1 - \frac{x^2}{t^2}\right)^{3/4} \tag{14}$$

$$\phi(x,t;c) = c\, e^{-(1-c)t}\, \phi(cx, ct; 1) \tag{15}$$

3.2 The diffusion Green's function

Written for the flux $\phi = vn$ the diffusion equation carries $1/v$ on the time derivative, since D is a length and Σ_a an inverse length, making both other terms ϕ/length:

$$\frac{1}{v}\frac{\partial \phi}{\partial t} - D\frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = \delta(x)\delta(t), \qquad \Sigma_a = (1-c)\Sigma_t, \qquad D = D_0(c)/\Sigma_t \tag{16}$$

Multiply by v. Integrating over $t \in (0^-, 0^+)$ leaves only the time derivative, so the pulse becomes an initial condition and the equation is source-free afterwards; absorption is uniform in space, so it factors out under $\phi = e^{-v\Sigma_a t}u$:

$$\frac{\partial u}{\partial t} = Dv\frac{\partial^2 u}{\partial x^2}, \qquad u(x, 0^+) = v\,\delta(x) \tag{17}$$

Fourier transforming, $\hat{u}_t = -Dvk^2\hat{u}$ with $\hat{u}(k, 0^+) = v$, so $\hat{u} = v\, e^{-Dvk^2 t}$, and inverting gives

$$\boxed{\phi_{\text{diff}}(x,t;c) = \frac{v\, e^{-(1-c)\Sigma_t v t}}{\sqrt{4\pi D v t}} \exp\left(-\frac{x^2}{4Dvt}\right)} \tag{18}$$

With $\Sigma_t = v = 1$ the two approximations differ only through D_0, which is $\frac{1}{3}$ for classical diffusion and $(1-c)\nu_0^2$ for asymptotic diffusion:

$$\phi_{\text{cl}}(x,t;c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi t/3}}\, e^{-3x^2/(4t)} \tag{19}$$

$$\phi_{\text{as}}(x,t;c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi(1-c)\nu_0^2\, t}}\, e^{-x^2/\left(4(1-c)\nu_0^2 t\right)} \tag{20}$$

where $c\nu_0 \operatorname{arctanh}(1/\nu_0) = 1$. For $c > 1$ the eigenvalue is imaginary, $\nu_0 = i|\nu_0|$, and $(1-c)\nu_0^2 = (c-1)|\nu_0|^2 > 0$; both branches tend to $\frac{1}{3}$ as $c \to 1$. Integrating (**??**) over all t recovers the steady solutions $\frac{\sqrt{3}}{2\sqrt{1-c}}e^{-\sqrt{3(1-c)}\Sigma_t|x|}$ and $e^{-\Sigma_t|x|/\nu_0}/\left(2(1-c)\nu_0\right)$, and $\int \phi\, dx = ve^{-(1-c)\Sigma_t v t}$ matches the exact solution, so the figures compare shapes at equal particle number.

c	0.6	0.8	1.0	1.2	1.5
$D_0(c)$	0.48588	0.39629	0.33333	0.28717	0.23745

Table 1: The asymptotic diffusion coefficient; the classical value is $1/3$ for every c.

3.3 Parts 3(a) and 3(b)

Each figure fixes c and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line is the causal front $|x| = vt$, beyond which the exact solution vanishes; only $x \geq 0$ is plotted, all three solutions being even in x.

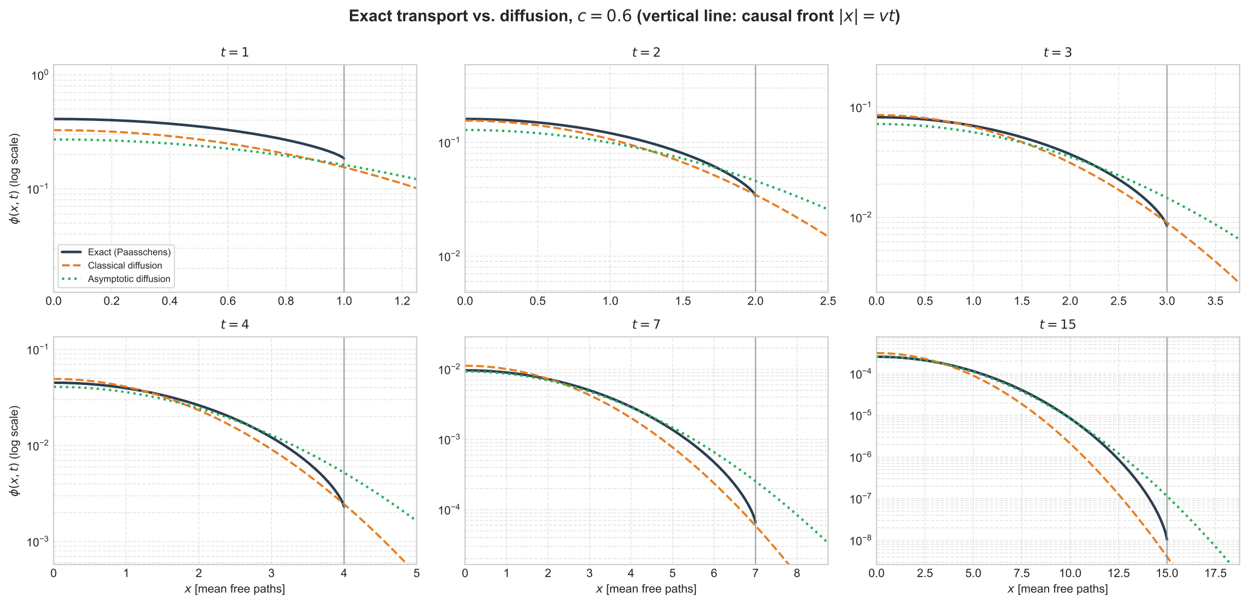

Figure 1: $c = 0.6$.

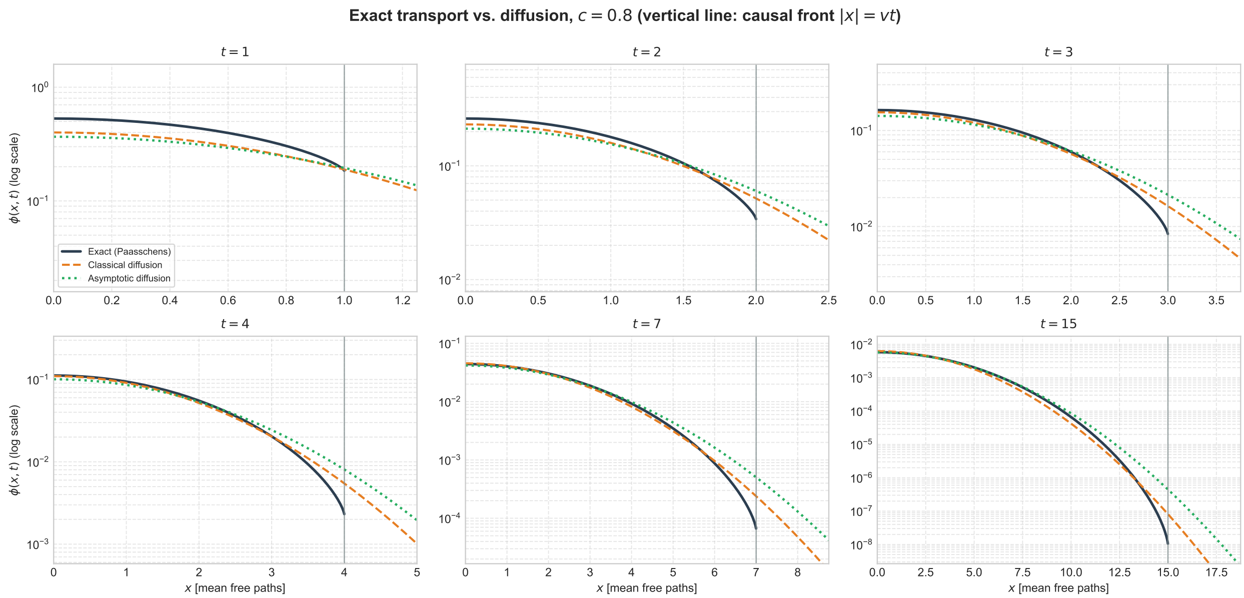

Figure 2: $c = 0.8$.

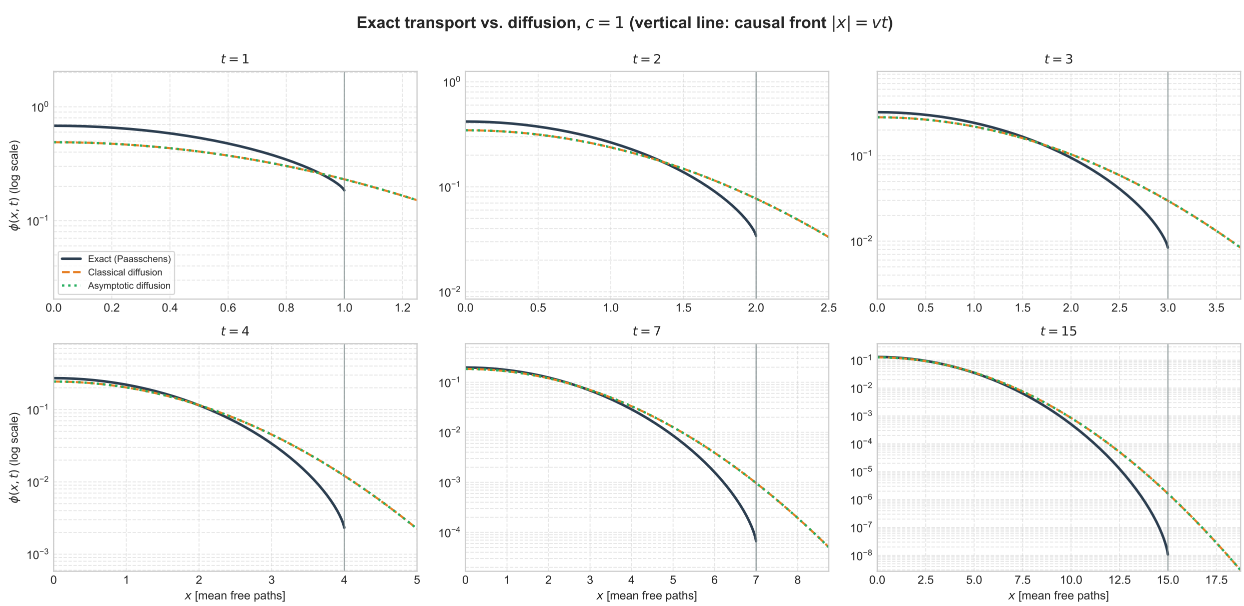

Figure 3: $c = 1$.

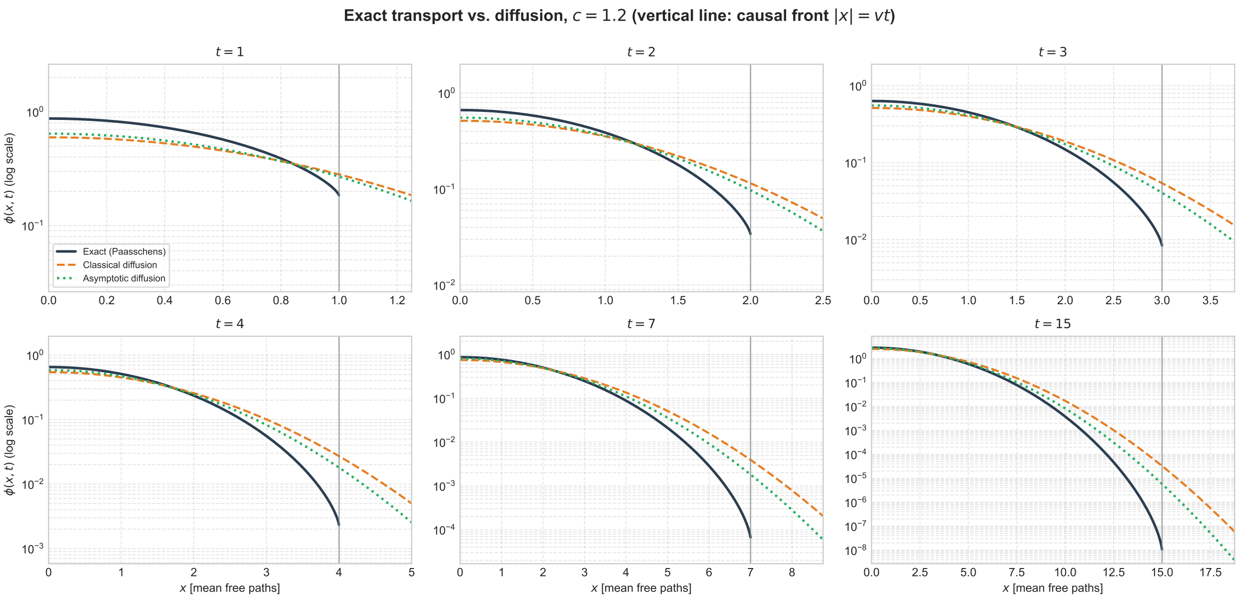

Figure 4: $c = 1.2$.

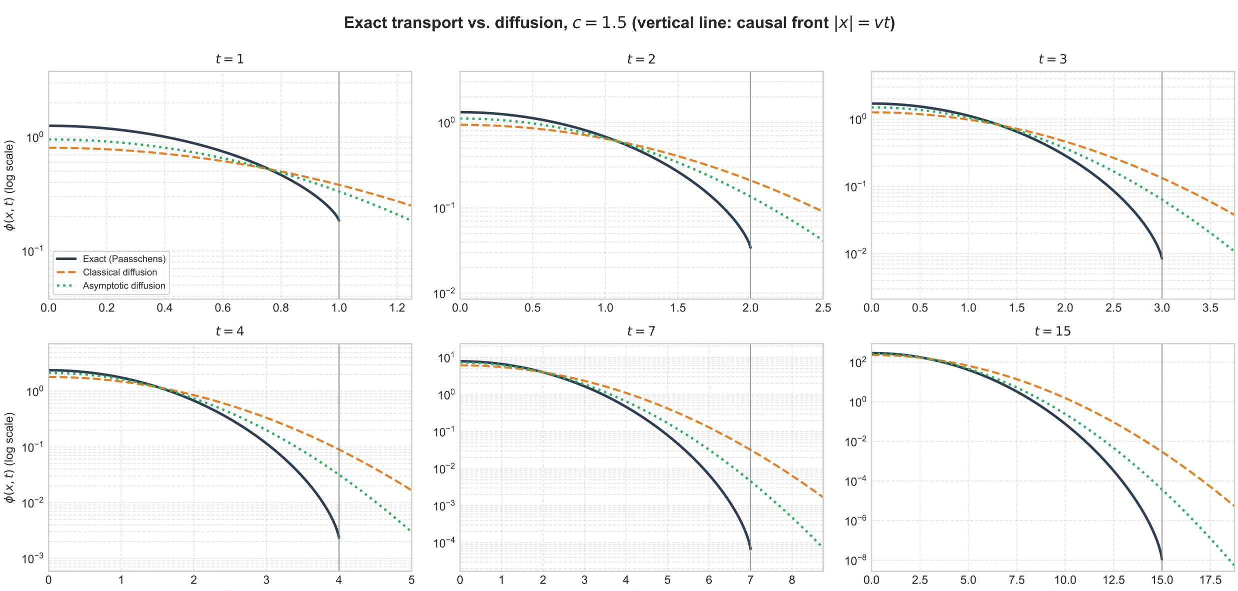

Figure 5: $c = 1.5$.

3.4 Part 3(c): the numerical solution

Solved in `src/homework2/solver.py` on the half domain $[0, L]$: centred second differences with ghost-node rows at both ends ($u_{-1} = u_1$ by symmetry, $u_{N+1} = u_{N-1}$ at a far L), the pulse as initial data $u_0 = v/h$, and Crank–Nicolson in time, $\left(I - \frac{\Delta t}{2} A\right) u^{n+1} = \left(I + \frac{\Delta t}{2} A\right) u^n$ with $A = \frac{Dv}{h^2}\mathrm{tridiag}(1, -2, 1)$, opened by four backward-Euler steps of $\Delta t/4$ — without which the scheme rings on the delta and returns a negative flux. Against (**??**) the largest relative error is 2.2×10^{-5} over all five c, both approximations and all six times; the particle balance is 1.00000000, and refining each discretisation with the other fixed gives error ratios of $4.04, 4.08, 4.33$ in space and $3.92, 4.07, 4.53$ in time, confirming second order in both. Below, the numerical solutions are drawn as markers over the analytic curves of the same colour.

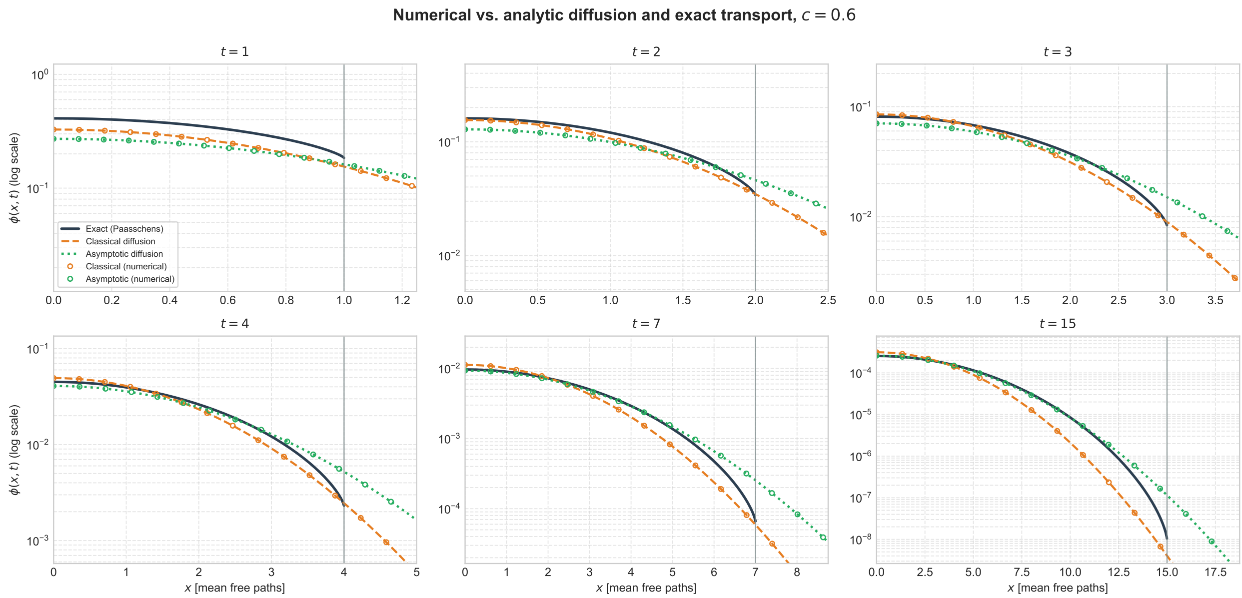

Figure 6: Numerical diffusion, $c = 0.6$.

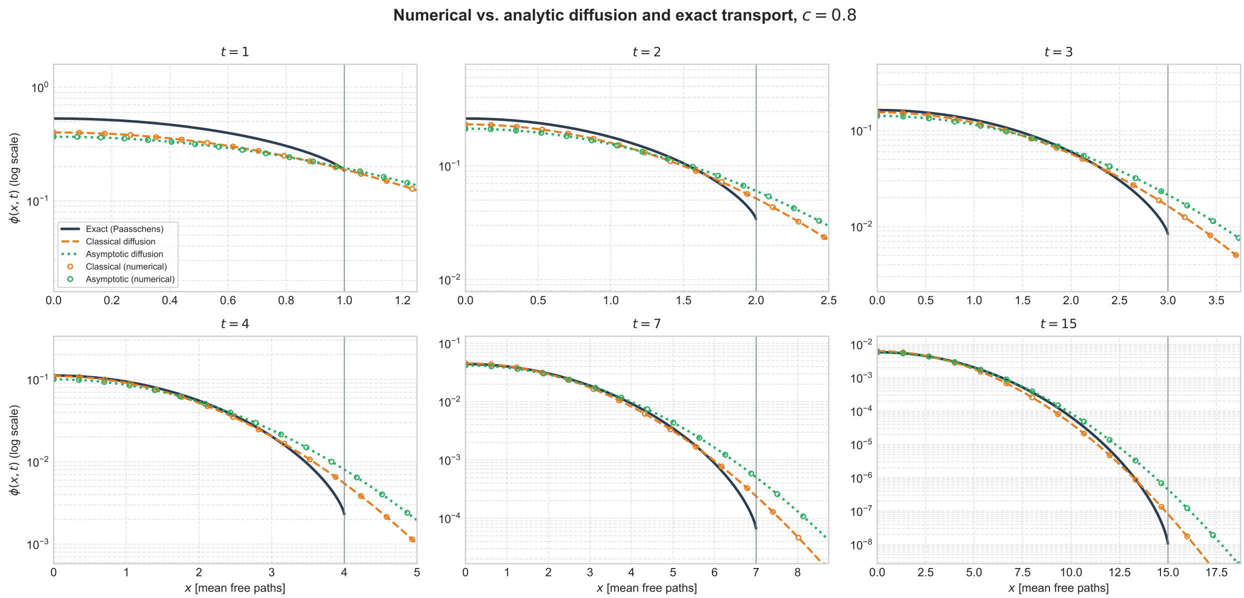

Figure 7: Numerical diffusion, $c = 0.8$.

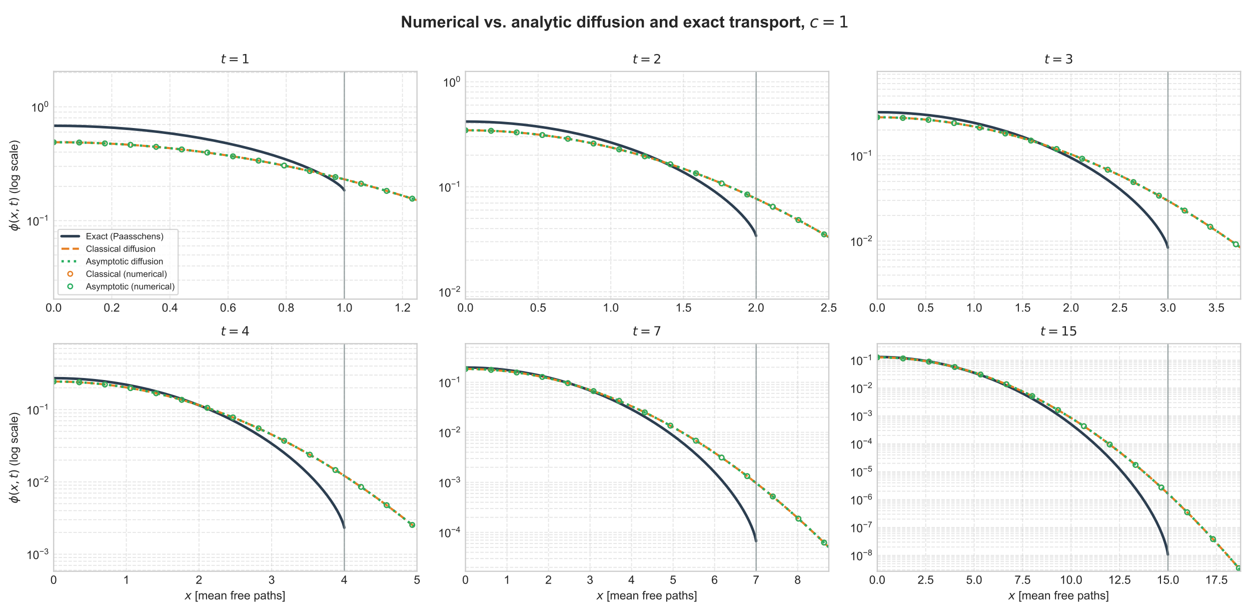

Figure 8: Numerical diffusion, $c = 1$.

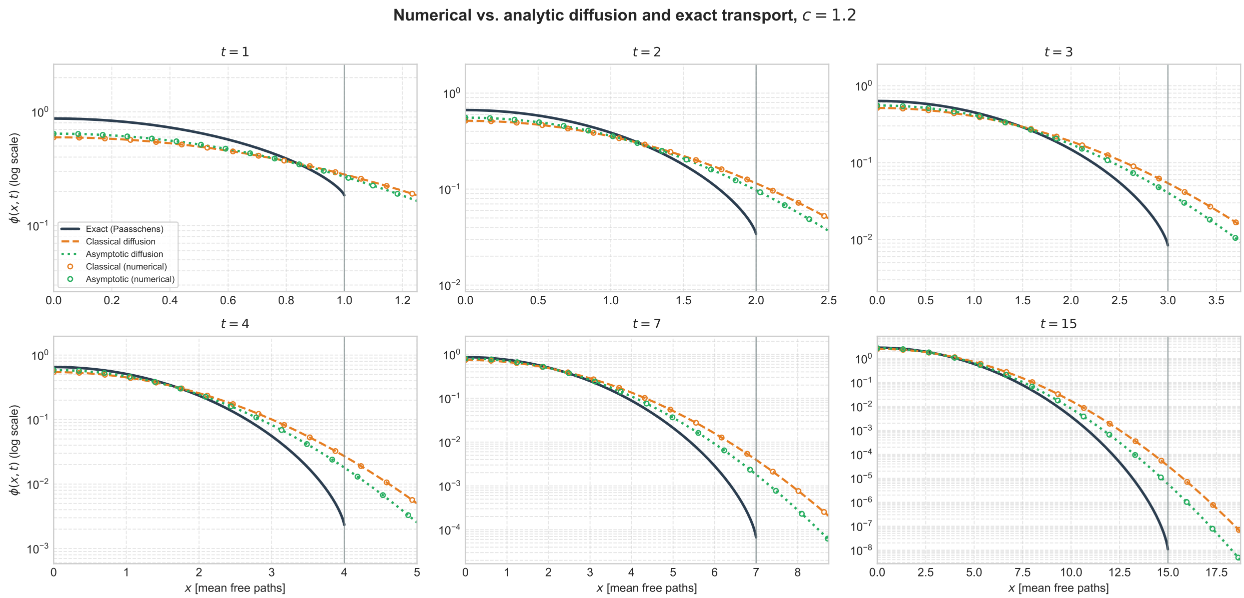

Figure 9: Numerical diffusion, $c = 1.2$.

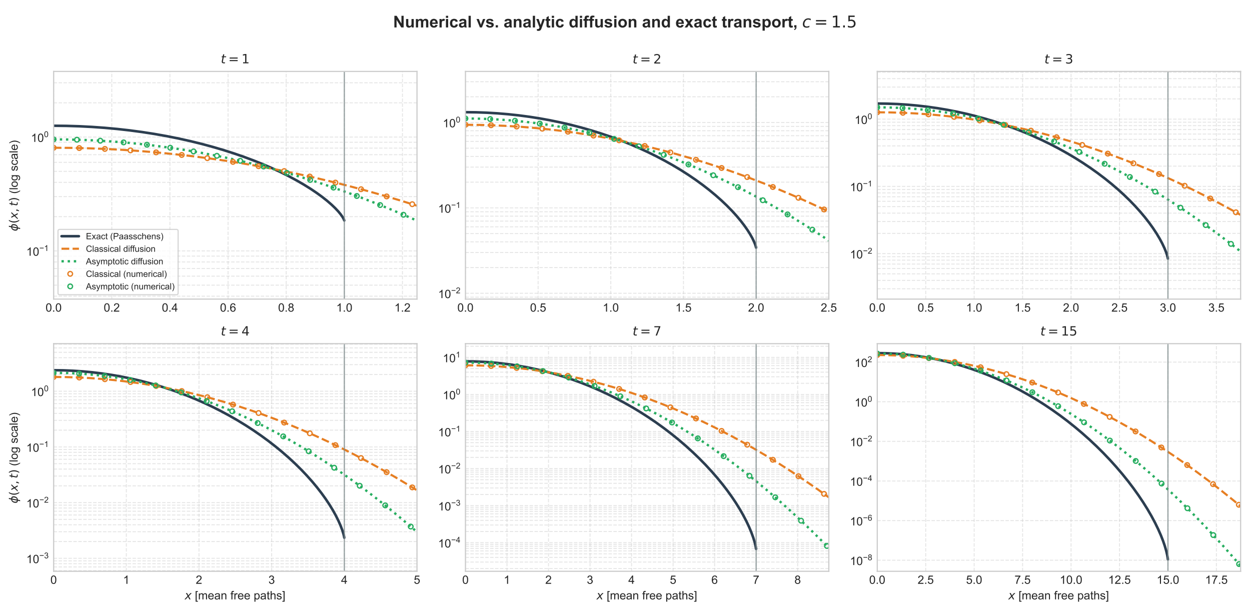

Figure 10: Numerical diffusion, $c = 1.5$.

3.5 Part 3(d): relative error against the exact solution

Relative error of each diffusion solution against the exact transport flux, analytic as lines and numerical as markers, evaluated out to $0.99\,vt$ — beyond the front the exact solution is zero and the relative error diverges for a reason that says nothing about the approximation. The downward spikes are sign changes, where a diffusion curve crosses the exact one. Three trends: the error falls with t as the angular flux isotropises; it grows towards the front, where diffusion's infinite propagation speed is worst; and asymptotic diffusion overtakes classical near the source once $|1-c|$ is appreciable, since it is built to carry the correct decay length, while at $c = 1$ the two coincide.

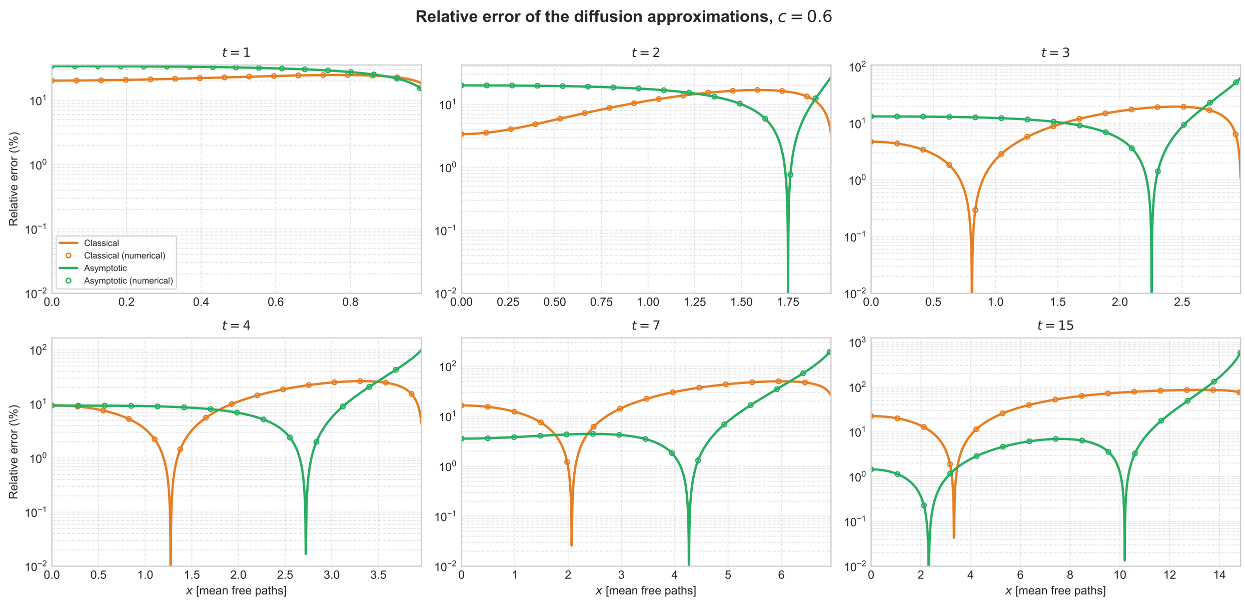

Figure 11: Relative error, $c = 0.6$.

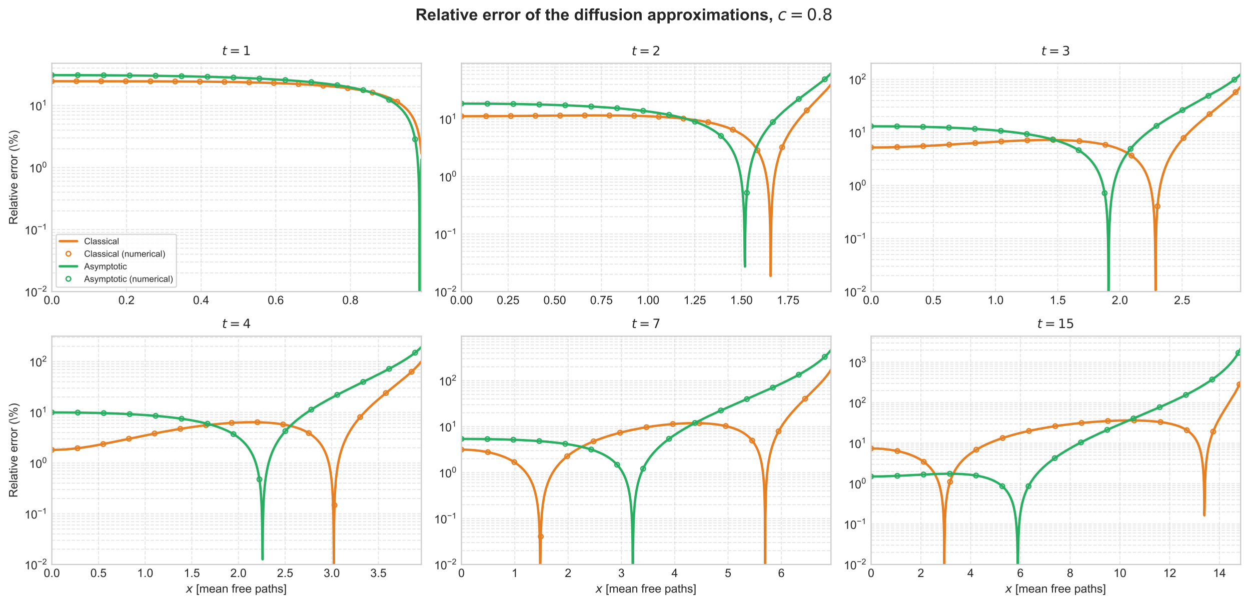

Figure 12: Relative error, $c = 0.8$.

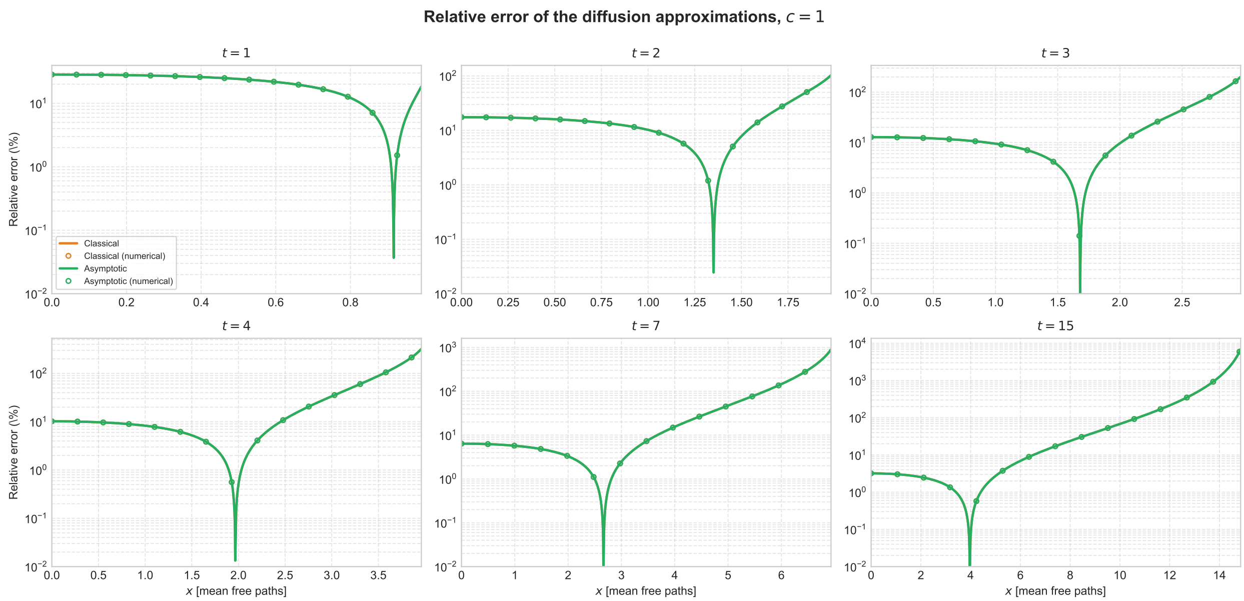

Figure 13: Relative error, $c = 1$.

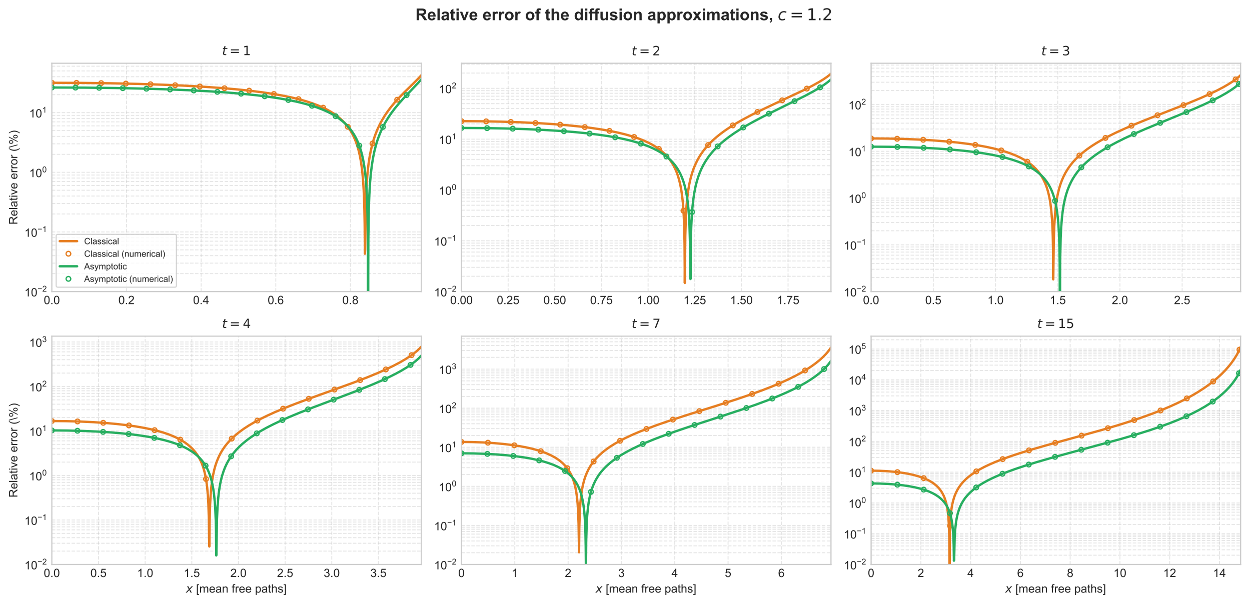

Figure 14: Relative error, $c = 1.2$.

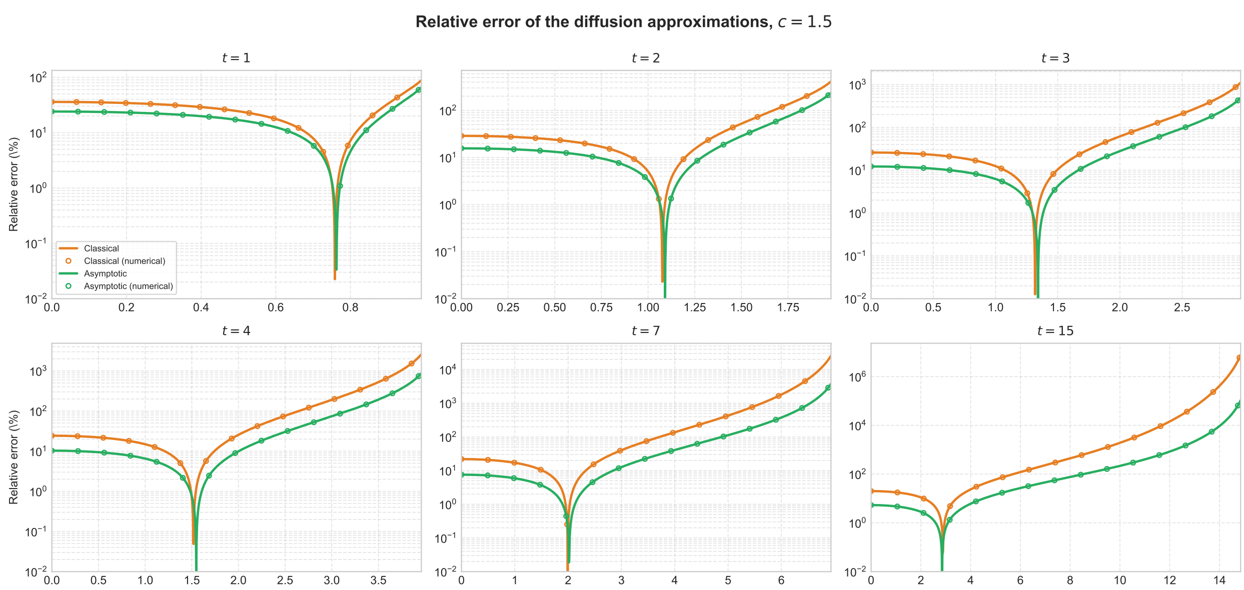

Figure 15: Relative error, $c = 1.5$.